CODESCHOOL · FOUNDATIONAL RESEARCH

Research Pack

When programming starts in Indian schools, a level-based design per grade, the market, CBSE's free CT&AI books, and the plan to validate it all with schools.

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Status: research complete; school interviews start 16 November 2026

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Interactive PDF: clickable contents, bookmarks, source links, fillable forms

How to use this PDF

This pack brings every research document from the CodeSchool repository into one place, with tools to act on it.

Feature	How to use it
Clickable contents	Every line on the contents page jumps to its section. Every page has a ↑ **Back to contents** link in the footer.
Bookmarks	Open the bookmarks or outline panel in your PDF reader (Acrobat, Edge, Chrome, Foxit) to move between sections and sub-sections.
Source links	Underlined teal text opens the official circular, syllabus, handbook or news source. Internal links jump within this PDF.
Fillable forms	The action checklist, hypothesis tracker, interview note form, device audit and B3 observation sheet have fields you can type in, tick and choose from. Save with **File → Save** in Acrobat Reader or Edge. Some browsers' built-in viewers don't save form data.
Print	Each form page prints cleanly. Print the B3 lesson and device audit pages for interviews.

Privacy: keep filled forms out of git

This repository is **public** on GitHub. Fill the interview forms in a **separate copy saved outside the repository**, with no real names. Never commit filled copies. The `.gitignore` already blocks `docs/research/interviews/` and `*.private.*` files, so name your copy `CodeSchool-Research-Pack.private.pdf`.

Source labels used throughout

Label	Meaning
Official	Checked against a board or government document (CBSE circular, syllabus, handbook, NEP).
News	Mainstream press.
Secondary	Publisher, vendor or blog source.
Background knowledge	Taken from well-known documents without re-checking in this research; verify before quoting.
Low-quality estimate	Paid market-research figures with undisclosed methods.
Hypothesis / recommendation	A design or pricing proposal to be tested, not a fact.

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1. Executive summary

The short answer, the evidence behind it, and what to do next.

The answer to the founding question

"Around 4th standard" is right about classrooms and wrong about policy. Children commonly meet Scratch around Class 3-4 in private-school books and Kerala's state books. Officially, India has four different starting points: unplugged computational thinking (CT) from the Foundational Stage (NEP) and from **Class 3** in CBSE (2026-27); block coding as a CBSE Class 3-5 competency; "coding activities" in Classes 6-8 (NEP); and examined text programming only from **Class 9 (ICSE, Java)** or **Class 11 (CBSE, Python)**.

Class 3
CBSE's CT&AI curriculum starts here in 2026-27 (50 h a year in Classes 3-5)

0 of 12
free CBSE handbooks contain any coding

120 h
of Classes 6-8 projects with no briefs or rubrics in any book

d ≈ 0.4
learning effect of game 'levels' in programming; badges and points show none

Rs 449 / 599
booklist pack prices to test (Classes 3-5 / 6-8); Uolo Tekie starts at Rs 799

Dec-Jan
school book lists close for the April session: the sales deadline

Key findings

- **Programming starts in layers, not at one class.** Thinking from Class 3 (CBSE), Scratch around Class 3-5 in practice, "coding activities" in Classes 6-8, and text programming from Class 9 (ICSE) or Class 11 (CBSE). CBSE's Classes 9-10 Computer Applications (165) has **no programming**. See section 3.
- **A level-based structure per grade is feasible and evidence-backed.** Levels are the one game mechanic with a measured learning effect. Recommended: five bands (A: Classes 1-2 to E: 11-12) of mastery-gated levels, separate from class, with placement tests and Express catch-up tracks. See section 4.
- **Schools, not parents, pay.** Parent-paid live coding collapsed (WhiteHat Jr; Tynker sold for ~$2.2M after a ~$200M purchase). School bundles survive (LEAD Group Rs 387 Cr FY26). Recommended model: **B2B2C**. See section 5.
- **CBSE gives away the puzzles but not the hard parts.** Across all 12 books: no coding, no project materials, no assessment templates; AI teaching text only in student books; hands-on AI only in Class 8. See section 7.
- **CBSE is telling schools coding isn't needed.** Its April 2026 webinar presents "no coding/software" as a design choice. Selling "CBSE requires coding" would contradict the Board. See section 7.12.
- **Best openings are what CBSE admits is unfinished:** the promised project list with rubrics and AI concept handouts are still not published, plus teacher support and assessment.
- **The biggest risks are policy churn and empty labs.** CBSE "advises" the curriculum and already deferred Class 9; only 64.7% of schools have computer facilities. See section 8.

Positioning in one line

CodeSchool is the easiest way to deliver the parts of CBSE CT&AI that CBSE didn't provide: project kits, AI teacher support, hands-on AI on low-end equipment, assessment and progress tracking, with coding as the "now you build it" tool. It sits on top of the free books instead of replacing them.

Top five next actions

#	Action	When
1	Fill the contact placeholders in the intro letter, print 20 B3 lessons and device audit sheets	9-11 Nov
2	Shortlist 25-30 schools matching the sample mix and send the first 10 intro letters	by 13 Nov
3	Run interviews with ~6 schools a week; fill the note form the same day	16 Nov - 11 Dec
4	Test the two new risks directly: coding demand, and whether CBSE's project list has arrived	every coordinator interview
5	Write the 2-page findings memo and make the pilot go/no-go call	14-18 Dec

The full, fillable list is in section 9: Action plan.

2. Scope and method

What was asked, what was covered, how, and what was left out.

The brief

Founder's request

"Before starting development, conduct market and curriculum research to determine the exact grade level at which programming is introduced in schools. While my initial understanding is around the 4th standard, I would like to verify this. Additionally, explore the feasibility of a level-based learning structure tailored to each grade. Do not start building or writing code yet; focus entirely on foundational research."

Research questions

#	Question	Answered in
Q1	At exactly which grade is programming formally introduced in Indian (and peer) school systems? Is "around Class 4" right?	Section 3
Q2	Is a level-based structure tailored to each grade feasible, and what does the evidence say about levels, mastery and progression?	Section 4
Q3	What does the market look like: players, failures, business models, pricing and size?	Section 5
Q4	Are the CBSE facts right? (added after the first report)	Section 6
Q5	What do CBSE's free CT&AI books actually contain, and what gaps remain? (added)	Section 7
Q6	How do we validate all this with schools before building? (added)	Section 10

In scope and out of scope

In scope

- India: NEP 2020, NCF-SE 2023, CBSE (fully verified), ICSE/CISCE, Kerala, Tamil Nadu, Karnataka, Atal Tinkering Labs, private textbooks
- Peers: England, USA, Finland, Australia, NZ, Estonia, Singapore, China, Korea, Japan, UAE
- Learning progressions and evidence on levels, mastery, PRIMM, Parsons, unplugged, block-to-text
- Market: Indian and global players, B2C vs B2B, pricing anchors, market-size claims
- All 12 CBSE CT&AI handbooks and the April 2026 webinar slides
- School interview plan and materials

Out of scope (not done)

- Any app design, code or prototype software (by instruction)
- Primary data: no schools, teachers or parents interviewed yet
- Maharashtra, AP/Telangana, Delhi boards (no reliable sources)
- CISCE Classes 1-8 guidelines, AI 417 / IT 402 syllabi, NCF-SE PDF (not read directly)
- Legal review of child-data rules (DPDP Act)
- A bottom-up market model from UDISE+ data

Method

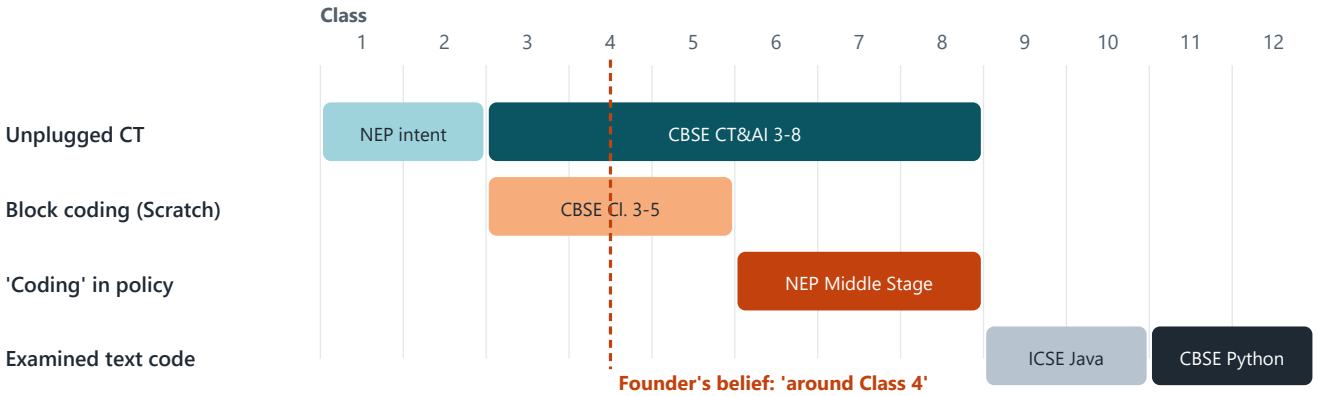
Stage	What was done	Output
1. Desk research	Parallel research on Indian curricula, global curricula, learning progression evidence and the market; synthesised into one report with labelled sources	Main report + 4 research notes
2. Interview plan	Hypotheses, sample, role guides, analysis and decision rule	Interview plan
3. Interview materials	Intro letter and consent script, Level B3 paper lesson, device audit, tracking sheet	Materials folder
4. CBSE verification	Full text of circulars Acad-15, -18, -70 of 2026, the CT&AI curriculum, and CA 165 / CS 083 / IP 065 syllabi	Verification note
5. Handbook review	All 12 student and teacher handbooks (Classes 3-8) and the 45-slide webinar deck	Handbook review

Limitations

- **No primary research yet.** Every conclusion about what schools want or will pay is a hypothesis until interviews are done.
- **Market size is unknown.** Published figures conflict by up to 12x and have no stated method; none should be used.
- **Uneven evidence.** No study ties specific coding concepts to specific ages; no block-to-text study covers ages 10-13; no Indian parent survey exists.
- **Handbook reading depth varies.** Front matter, activities and AI chapters were read in full in all books; Class 5 puzzle chapters were skimmed.
- **Corrections were made along the way.** Two CBSE claims in the first draft were wrong and one later correction was itself reversed; the record is kept in section 6.

3. When does programming start?

Four different answers depending on what "programming" means. Treating them as one causes the confusion.



Indian Class *n* is roughly England Year *n*+1. The dashed line marks the founder's starting belief.

The verdict, layer by layer

Layer	Earliest official start	Where children actually meet it	Is "around Class 4" right?
Unplugged CT	Foundational Stage, preschool to Class 2 (NEP intent); **Class 3** (CBSE, 2026-27, 50 h a year)	Class 3 CBSE handbooks; Kerala Class 3 logic games	**No. One class too late.**
Block coding (Scratch)	**Class 3-5** competency in CBSE's framework, where devices allow	Class 3 private textbooks; Kerala Class 4; ICSE-aligned books Classes 4-5	**Yes, roughly.** This is what the belief describes.
"Coding" in policy	**Classes 6-8** (NEP Middle Stage; CBSE CT&AI 6-8; optional modules since 2021)	Scratch games, HTML, BASIC-256 or intro Python in Class 6 textbooks	**No. Two classes too early.**
Graded text programming	**Class 9** elective in ICSE (Java); **Class 11** in CBSE (Python, CS 083 / IP 065)	Kerala Python in Classes 9-10; some Class 6 books preview Python	**No. Five to seven classes too early.**

Board-by-board status (November 2026)

Board / programme	Start	What is taught	Source
NEP 2020	CT from Foundational Stage; coding activities Classes 6-8	Puzzles and games for CT (para 4.25); policy, not a syllabus	Official
CBSE CT&AI, Classes 3-8	**Class 3** (2026-27)	3-5: maths-puzzle CT, 50 h in Maths and TWAU; 6-8: 100 h (40 CT, 20 AI, 40 projects). Schools "advised"; Acad-18 says it "mandates" content	Acad-15, curriculum
CBSE Class 9 CT&AI	**Deferred** for 2026-27	Acad-70 (29 Oct 2026): modules "shall not be the part of the curriculum for these batches"	Acad-70
CBSE Classes 9-12 electives	CA 165 (Cl. 9-10); CS 083 / IP 065 (Cl. 11-12)	CA 165: office tools and HTML, **no programming**; CS / IP: **Python**, SQL, Pandas	Official syllabi
CISCE (ICSE) 1-8	No official class	Publishers decide; Scratch around Classes 3-5	Secondary
ICSE 9-10	Class 9 elective	Java with BlueJ; Robotics & AI	Secondary
Kerala (KITE)	Class 3 logic; **Class 4 Scratch**	Games Class 6, AI Class 7, Python 9-10, robotics Class 10	News
Tamil Nadu	Class 6	TN SPARK pilot in ~5,000 government schools	News
Karnataka	Class 9 (AI vocational)	Class 6 under study	Secondary
Atal Tinkering Labs	Class 6	~10,000 labs; tinkering, not a curriculum	News
Private textbooks	Class 3 Scratch	Scratch Classes 3-6; HTML or intro Python in Class 6	Secondary

Two caveats. (1) It is unclear how mandatory CBSE's Class 3-8 curriculum is: "advised" in Acad-15, "mandates" in Acad-18, but no board exam or enforcement. (2) Devices limit everything: UDISE+ 2024-25 shows only **64.7%** of schools with computer facilities (Bihar government schools 16.5%).

International comparison

Country	CT / algorithms start	Blocks	Text language	Mandatory?
England	Age 5 (KS1)	Ages 7-11 (KS2)	Ages 11-14, at least one textual (KS3)	Statutory (maintained schools)
USA (CSTA 2026)	PK/K	Elementary (Code.org CSF)	Middle/high school	Voluntary; 12 states require CS to graduate
Finland	Grade 1, age 7	Grades 3-6	Grades 7-9 in maths	Yes, embedded
Australia	Foundation, age 5	Years 3-6	Years 7-8	To Year 8
Singapore	P4-P6 (Code for Fun)	10-hour CFF in all primaries	O-level Python (elective)	CFF yes; AI for Fun opt-in
China	~Grade 3 (IT subject)	Primary graphical	Python in junior/senior high	IT compulsory; Beijing 8 AI h/yr
South Korea	Grades 5-6	Elementary informatics	Middle school	Yes
UAE	Kindergarten	n/d	n/d	Government schools, 2025-26
India	Class 3 formally (~age 8)	Class 3-5 competency	ICSE Cl. 9; CBSE Cl. 11	"Advised"; programming elective

Pattern everywhere: unplugged at 5-7, blocks at 7-11, text at 11-14, exam CS at 14-16. AI is being layered on top of CS, and in places reversed (Korea downgraded AI textbooks within six months). Several peer rows are background knowledge; verify before quoting externally.

4. A level-based structure per grade

Feasible and evidence-backed: five bands of mastery-gated levels that are separate from the child's class.

- **Levels work.** Costa's 2023 meta-analysis: gamification gain in programming of **d ≈ 0.4**, largely from levels; badges, points, avatars and leaderboards had no significant learning effect.
- **Frameworks already use bands.** CSTA (multi-year levels), Bebras (two-year age groups) and Code.org's A-F courses with "Express" catch-up versions.
- **Developmental seams** fall around age 7-8 (structured → open blocks) and 10-12 (concrete → abstract), not at every grade.

Recommended framework (recommendation)

Band	Levels	Concepts and tools	Exit outcome	Alignment
A: Explorers Classes 1-2	A1 sequencing · A2 debugging · A3 repeat · A4 events · A5 story project	Unplugged first, then icon blocks in 10-20 min teacher-led sessions; story framing	Follows and gives precise instructions; fixes one bug; makes a short story	NEP Foundational intent; no CBSE mandate (cheap, mostly printable)
B: Builders Classes 3-5	B1 loops · B2 events · B3 conditionals · B4 nested/while · B5 variables · B6 CT puzzles · B7 capstone	Unplugged intro then Scratch-style blocks; Android-first, offline; printable sheets	Builds an interactive game with events, conditionals and a loop; explains it	**CBSE CT&AI 3-5** (50 h); needs a strand mapped to the NCERT maths chapters
C: Makers Classes 6-8	C1 functions · C2 lists/data · C3 AI literacy I · C4 Python behind blocks · C5 reading Python · C6 writing Python · C7 AI literacy II · C8 project	Blocks and text together, then text; Parsons scaffolds; no-code AI tools	Multi-part project; reads and modifies Python; trains and critiques a classifier	**CBSE CT&AI 6-8** (100 h); NEP coding activities; block-to-text bridge
D: Coders Classes 9-10	D1-D6: Python consolidation, modular code, data structures, algorithms, data/AI project, testing AI-written code	Text; Python track (CBSE/state) and Java track (ICSE)	Writes and tests modular programs; evaluates AI code	No CBSE programming subject here: the Scratch-to-Python gap
E: Specialists Classes 11-12	E1-E6 exam tracks plus specialties	Text; portfolios; exam practice	Board-exam readiness and a portfolio	CBSE CS 083 / IP 065, ISC CS; separate later business

How levels decouple from class

- **Default entry level and an age-appropriate "skin"** for each class; the level comes from what the child can show.
- **10-15 minute adaptive placement check** on joining; **"challenge to skip"** tests at each level boundary.
- **Mastery gating** at ~80-90% of check items, with spaced review.
- **Teacher override** so a class can stay together on a period plan (overrides recorded, remediation queued).
- **Express tracks:** Junior (Classes 4-5 newcomers cover Band A fast), Middle (Classes 6-8 beginners cover A-B in a term), Senior (Class 9 beginners straight into Python/Java).

One lesson template a non-specialist can run

1 Predict & run	2 Investigate	3 Modify	4 Parsons	5 Make
worked example	tracing questions	guided change	reorder blocks	open-ended

PRIMM with a Parsons bridge. Steps 1-4 can be auto-graded; step 5 protects against puzzle-only progression. A worked example is the Level B3 lesson.

Evidence behind each design choice

Design choice	What the evidence says	Strength
Levels as the core mechanic	d ≈ 0.4 in programming, driven by levels; badges/points/leaderboards no significant effect	Moderate
Badges optional, private, mastery-based	Badges and leaderboards lowered motivation and scores in one study; game elements raise quantity not quality	Moderate (contested)
Mastery gating	Direction strong; realistic effects much smaller than Bloom's "2 sigma"	Strong direction; size contested
PRIMM lessons	UK quasi-experiment, ages 11-14: PRIMM group scored higher	Moderate; nothing below age 11
Parsons problems	Same gains as writing code, in less time	Moderate; one null study
Data-driven hints	iSnap next-step hints; explaining "why" helps	Moderate
Block-to-text around Classes 7-8	Blocks faster for concepts; advantage fades ~10 weeks after Java, no harm	Moderate; no study at ages 10-13
Unplugged then plugged	Large unplugged effects but plateaus; hands-on beat unplugged-only	Moderate
Story-based tools for Classes 1-2	d = 0.81 programming → CT at ages 3-7; structured tools best	Moderate
Screen time	No hard cap for ages 6-8 in any guideline	Weak; a design decision
Scripted support for teachers	Mostly satisfaction data; teacher readiness moderates outcomes	Weak

5. Market

Parent-paid live coding burned out; school bundles survived. Free basics push value into alignment, assessment and teacher support.

B2C collapsed, B2B survived

Player	Model	What happened	Source label
WhiteHat Jr	B2C live 1:1 classes	~150,000 paid students at 2020 peak; 26.73% of BYJU's FY21 loss; layoffs, lawsuits, refund complaints	News / user posts
Tynker (BYJU's)	B2C app + schools	Bought for ~$200M, sold for ~$2.2M (2025)	News
BrightChamps	B2C live classes	FY23 revenue Rs 20.6 Cr vs loss Rs 159.5 Cr	News
LEAD Group	**B2B school bundle**	FY26 revenue **Rs 387 Cr**; EBITDA breakeven; Rs 415 Cr ARR	News
Eupheus / Extramarks	B2B school bundles	~Rs 225 Cr / Rs 233 Cr FY24	News
Codingal / PlanetSpark	B2C live	Good ratings and growth, but teachers cost Rs 700-2,833 per class	News / secondary

Recommended model: B2B2C. Sell a per-student annual licence to private CBSE and ICSE schools (collected in fees), add a free parent view and an optional paid home-practice tier. Government (B2G) is a second stage through ATL and ICT-lab vendors on GeM. Book lists close **December-January** for an April start.

Where the paid value has to sit

Basic block coding is free (Code.org, Scratch, CBSE books). Schools pay for what they struggle with:

- **Syllabus mapping** to CBSE CT&AI, CISCE and state syllabi (by NCERT maths chapter).
- **Assessment and reporting** for an internally assessed subject.
- **Support for non-specialist teachers:** period scripts, answer keys, misconception notes, a "who is stuck" dashboard.
- **Android-first, offline delivery** (90% of rural teens have a phone at home; ~31% own one).
- **Regional languages.**
- **A structured block-to-text bridge** for Classes 6-8, which neither CBSE nor Scratch provides.

Positioning and pricing hypotheses

Element	Hypothesis	Confidence
Core offer (Classes 3-8), revised 6 Nov 2026	**Rs 449 (Classes 3-5) and Rs 599 (Classes 6-8) booklist packs**, or a **Rs 299 + GST** school licence for premium schools; was Rs 200-800	Hypothesis; printed CT&AI books sell at Rs 179-630, Uolo Tekie from Rs 799 (competitors report)
Classes 1-2 (Band A)	Free or nominal inside the school licence	Hypothesis
Classes 9-12 exam tracks	Separate add-on priced above the core	Hypothesis
Parent tier	Free progress view; paid home practice billed monthly, no EMI, published refund policy; test below Rs 500/month	Hypothesis
B2G	Software + teacher-training layer alongside ATL / ICT vendors	Hypothesis

Market size is unknown; do not quote published figures. "$14B coding for kids in India" has no source. Global estimates range $514M-$6.5B; Indian K-12 estimates differ 2x. Illustration only: 5M private-school students × Rs 500-1,500 = Rs 250-750 Cr a year across the market. The real model is schools × Class 3-8 enrolment × realised price, from UDISE+ and pilot conversion.

Pitch shift: AI literacy is replacing "learn to code". Code.org rebranded Hour of Code as Hour of AI (Dec 2025); CBSE named its mandate "Computational Thinking and AI". Avoid outcome promises (ASCI found 5 of 7 WhiteHat Jr ads in potential violation).

6. CBSE facts checked against official documents

Every CBSE claim in the report was rechecked against the full official text on 5 November 2026.

Claim	Verdict	What the documents say
CT&AI for Classes 3-8 starts in 2026-27	**Confirmed**	Acad-15/2026, 1 April 2026
Schools "advised", not ordered	**Confirmed, with nuance**	Acad-15 "advised"; Acad-18 "mandates" content and schools "should" use the free books
50 h (3-5); 100 h (6-8: 40 CT + 20 AI + 40 projects)	**Confirmed**	Curriculum sections 3.4 and 10.2
Classes 3-5 embedded in Maths and TWAU	**Confirmed**	Books follow the NCERT maths chapter order
Scratch named for Classes 3-5	**Confirmed**	Competency C-6, a single competency
Teachable Machine named for Classes 6-8	**Right for Class 8 books; not in the curriculum**	Curriculum names no tool. Class 8 student book names Teachable Machine and Machine Learning for Kids with three labs; Classes 6-7 name none
Python not a Class 3-8 topic	**Confirmed**	Appears only in the "Aims"
Class 9 CT&AI dropped for 2026-27 batch	**Confirmed**	Acad-70/2026, 29 Oct 2026
Compulsory Class 9 in 2027-28; Class 10 exam 2028-29	**Still unconfirmed**	Neither circular says this
CA 165 teaches Python/HTML in Classes 9-10	**Wrong**	No programming: office tools (9), HTML/CSS (10)
CBSE's first programming subject is in Class 11	**Confirmed (new)**	Python in CS 083 and IP 065
Teacher training through NISHTHA	**Not supported; CBSE runs its own CPD**	Webinar: 2026-27 theme "Computational Thinking and Understanding AI"; district/regional 1-day workshops, lead-school half-days, DD PM e-Vidya videos

What this changes for CodeSchool

- **No CBSE student meets a programming language before Class 11.** Band C (block-to-text) and Band D sit in that gap.
- **Classes 3-5 CT is maths puzzles in NCERT chapter order**, so Band B needs a strand mapped to those chapters.
- **Explicit if-then is a Class 7 outcome**, but if-then rules appear in puzzles from Class 3, so Level B3 at Class 4 is a stretch, not a mismatch.
- **Classes 6-8 AI is taught by computer teachers**: they are buyers for Band C, not just a device audit.
- **Platform-agnostic and free-first**: no incumbent tool, and schools are steered to free options; sell tooling and assessment, not content.

> **Correction history.** The first draft said CA 165 teaches programming (wrong) and named Teachable Machine for Classes 6-8. A later "correction" said no tool was named anywhere; reading the Class 8 student book showed the original notes were right about the handbooks. The verification table above is the final position.

7. CBSE CT&AI handbook review

All 12 free books (student and teacher, Classes 3-8) plus the 28 April 2026 webinar. These are what every school will compare CodeSchool with.

7.1 What was read

Class	Student Handbook	Teacher Handbook
Class 3	53 pages	96 pages
Class 4	52 pages	111 pages
Class 5	61 pages	118 pages
Class 6	84 pages	110 pages
Class 7	86 pages	98 pages
Class 8	75 pages	92 pages

- **Read in full in every book:** front matter (goals, outcomes, hours, pedagogy, assessment), every activity, every AI chapter.
- **Read question by question:** puzzle chapters of Class 3 TH, Class 4 SH and Classes 6-8 TH; AI chapters of Classes 6-8 SH.
- **Skimmed:** Class 5 puzzles and the other student-book puzzle chapters (same questions as the teacher books, without solutions).
- **Made by:** Centre for Creative Learning, IIT Gandhinagar, with experts from IIT Madras, Azim Premji University, MNIT and NITTTR. Activities cite CS Unplugged.

7.2 Coverage at a glance

	Class 3	Class 4	Class 5	Class 6	Class 7	Class 8
Puzzle bank by maths chapter	Provided	Provided	Provided	Provided	Provided	Provided
Unplugged activities	Thin	Thin	Thin	Thin	Thin	Thin
Coding / Scratch	Missing	Missing	Missing	Missing	Missing	Missing
AI teaching text (student book)	n/a	n/a	n/a	Provided	Provided	Provided
AI support in teacher book	n/a	n/a	n/a	Thin	Thin	Thin
Hands-on AI	n/a	n/a	n/a	Missing	Missing	Provided
Project briefs & rubrics	n/a	n/a	n/a	Missing	Missing	Missing
Assessment templates	Missing	Missing	Missing	Missing	Missing	Missing

Class 8 hands-on AI = three Teachable Machine labs (need computer, internet, webcam, microphone).

7.3 Book by book

Class	CT chapters	Unplugged activities	AI part	Coding	Projects
3	14	**Caesar cipher** with a paper cipher wheel	None	None	n/a
4	14	**Coin Flip Magic** (parity, error detection); floor **Sorting Network**	None	None	n/a
5	15	**Searching with a weighing scale** (linear vs divide-and-conquer)	None	None	n/a
6	10	12-point circle step rule	4 chapters: AI basics incl. supervised/unsupervised/reinforcement learning, data, patterns, digital ethics. No hands-on	None	**None**
7	8	Punched-card sorting; logic punch cards	4 chapters: domains, classification/regression/clustering, case studies, charts, bias. No hands-on	None	**None**
8	7	Height in decimal/binary/ternary "pipes"	4 chapters + **3 Teachable Machine labs**	None	**None**

	Student Handbook	Teacher Handbook
Puzzles	Questions only	Same questions with worked solutions
Activities	Worksheet pages	Lesson plan with timings (30-45 min), template/worksheet/video links, CT skills
AI chapters (6-8)	**Teaching text** with examples, then exercises	**Exercises and answer keys only**

Advanced CT 40 h	**Intro AI 20 h**	**Interdisciplinary projects 40 h (no material in books)**

Classes 6-8: 100 hours a year each (Classes 3-5: 50 h embedded in Maths and TWAU)

7.4 Finding 1: built around the maths lesson

Every book is "a companion to the Mathematics textbook" and "follows the same chapter sequence". Chapter titles match the new NCERT maths books one for one (e.g. Class 3: *What's in a Name? … The Surajkund Fair*; Class 8: *A Square and a Cube … Proportional Reasoning*).

> **So what for CodeSchool**
>
> Index everything CodeSchool sells for Classes 3-8 by **NCERT maths chapter** so a teacher finds "today's" material in one tap.

7.5 Finding 2: the puzzles are hard, even in Class 3

Class 3 (age 8) includes finding the heavier of 12 balls in the fewest weighings, data-sufficiency questions and multi-clue passcodes. Student books give no hints or answers and no easier-to-harder ramp.

> **So what for CodeSchool**
>
> Levels add real value: warm-ups, hints and worked examples for the same chapter skills, with harder levels for those ready.

7.6 Finding 3: rules, branching and search from Class 3, always as puzzles

Class 3: "When on a WHITE block, he must skip the next TWO blocks. When on a BLACK block, … the next ONE" (an if-then rule); a frog that jumps 4 and slips 2 (a loop); path tiles from Home. Class 4: a rule with an exception; tracing a button sequence. Class 5: linear search vs divide-and-conquer. Students only **follow** or **compare** algorithms, never **write** them.

> **So what for CodeSchool**
>
> Band B levels as "**the puzzle in this chapter, now you build it**". Level B3 fits the Class 3 skip rule and Class 4 *Hide and Seek / The Cleanest Village*.

7.7 Finding 4: no coding in any book, and CBSE says that's intended

Scratch is a listed Classes 3-5 competency and CG-3 covers computers, hardware and software, but none of the 12 books has content for either. The webinar says "No coding/software", "no 'write a program'", and for Class 6 "Recognise — not build".

> **So what for CodeSchool**
>
> The coding gap is real on paper but schools may not feel it. Sell coding as a "now you build it" extension or via parents, not as a CBSE requirement. **Test in interviews** (coordinator Q9, H10).

7.8 Finding 5: 40% of Middle Stage time has no materials

Each of Classes 6-8 gives 40 h to "two required projects". No brief, rubric or example in any book, though the Class 6 book tells teachers to use "clear rubrics". The webinar gives one topic per grade (Class 6 weather across states; Class 7 "Nutri-Quest" meal planner; Class 8 sports data) and promises an "AI Project list with rubrics" as coming soon: still not on the CT&AI page six months later.

> **So what for CodeSchool**
>
> **Project kits for Classes 6-8** cover 120 hours across three years. The window may close when CBSE publishes its list: check monthly and align to its topics.

7.9 Finding 6: AI is taught in student books; only Class 8 is hands-on

Student books explain AI in plain text with Indian examples (Turing Test, ML types, classification/regression/clustering, ICMR TB X-ray screening, Pune waste app). Teacher books carry only exercises and answer keys. Classes 6-7 have no hands-on AI. Class 8 has three Teachable Machine labs: a waste classifier (photos of plastic/paper/organic), voice commands (Start/Stop/Help) and a cat/dog bias test (dataset "in the Drive", no link in the text). Each group needs a computer, internet, webcam and microphone. The Class 8 text uses a six-stage lifecycle while its outcome uses four stages.

> **So what for CodeSchool**
>
> **Teacher scripts and slides** for AI chapters; **hands-on AI for Classes 6-7**; a **device-light, projector version** of the Class 8 labs with datasets supplied.

7.10 Finding 7: few, borrowed, online-dependent activities

Eight unplugged CT activities across six classes, mostly adapted from CS Unplugged, relying on Google Docs, YouTube and tinyurl links; the sorting network has to be chalked on the floor.

> **So what for CodeSchool**
>
> A printable, offline activity kit with a teacher script for **every** chapter.

7.11 Finding 8: continuous assessment, nothing to track it

Every book lists written CT puzzle tests, group activities and a **Teacher Observation Journal**; none gives a template, rubric or record.

> **So what for CodeSchool**
>
> A simple observation journal and progress record mapped to CBSE competencies, printable for report cards.

7.12 What the CBSE webinar adds (28 April 2026)

Source: webinar slides (45 slides, read in full); recording not yet watched.

Topic	What the slides say	Implication
Coding	"No coding/software"; "no 'write a program'… no expectation of computer access for assessment"; "largely unplugged"	Don't pitch "CBSE requires coding"
AI depth	Class 6: "Recognise — not build… don't need to write code or train models"	Class 8 labs go further than this
Projects	One data-project topic per grade; "AI Project list with rubrics" coming soon	Biggest open gap; time-limited
Coming soon	Dedicated CT&AI page (now live); AI "Concept Handouts"; project list with rubrics	Two of three still missing
Who teaches 6-8	CT through Maths, Science **and Social Sciences**; AI and projects "anchored by the Computer teacher"	More teachers to support
Training	Annual theme "Computational Thinking and Understanding AI": district 1-day (6 CPD h), lead-school half-day (3 h), online, DD PM e-Vidya CBSE 15, regional workshops	Short training → in-product guidance matters
Teacher book claim	"Everything in the student handbook, plus" solutions	Not true for the 6-8 AI chapters

7.13 Don't compete with the books: sit on top of them

CBSE gives (free)	CodeSchool could give (paid)
Puzzle banks by maths chapter, Classes 3-8	The same skills as **levels** with hints, on-ramps and auto-checked practice, on phone, projector or paper
No coding in any class	**Block-coding levels** tied to each chapter's puzzles ("now you build it")
8 unplugged activities across six classes	An activity for every chapter, printable kits and a teacher script
No project materials	**Project kits for Classes 6-8**: briefs, rubrics, examples, assessment
AI text in student books; answer keys for teachers	**Teacher scripts and slides** for every AI chapter
Hands-on AI only in Class 8, needing devices and internet	**Hands-on AI for 6-7** and a device-light Class 8 lab version
"Teacher Observation Journal" with no template	A **progress journal and dashboard** mapped to CBSE competencies

8. Risks and open questions

Three assumptions could fail: the CBSE curriculum holds, schools have devices, and principals pay for something CBSE gives away free.

Risk	Why it matters	Mitigation
Policy volatility	CBSE "advises" Classes 3-8 and deferred Class 9; Korea reversed AI textbooks	Concept-based framework; board mapping as a swappable layer
CBSE says no coding (new)	Webinar presents no-coding as a design choice	Lead with projects, AI support and assessment; coding as extension
CBSE fills the project gap (new)	Project list with rubrics promised	Ship fast; align kits to CBSE topics; monitor monthly
Devices and labs	64.7% have computer facilities; many sanctioned labs not working	Unplugged-first, printable kits, projector mode, offline Android
Free substitutes	Code.org, Scratch, CBSE books	Sell alignment, assessment, teacher tooling, languages
Teacher capacity	Non-specialist teachers; one-day CBSE training	Scripted PRIMM template, micro-videos, auto-grading, dashboard
Shallow puzzle progression	Points raise quantity, not quality	A make step in every level; capstones
Advancing without mastery	Pacing pressure undermined Cognitive Tutor	Mastery gates; logged overrides
Trust	WhiteHat Jr ads, suits and refunds damaged the sector	Monthly billing, refund policy, no outcome claims
Sales cycle	Book lists close Dec-Jan	Pilot-ready product and materials by ~September
Children's data	Data on minors	Legal review of DPDP child-consent rules before launch

Open questions and how to answer them

Open question	Ask whom	How to validate
Did schools implement CBSE CT&AI in 2026-27, with what?	Principals, coordinators	15-20 school interviews
Will principals pay alongside free books, and how much?	Principals, owners	Price anchors; LOIs before December
Can a maths teacher run a level in one period untrained?	Class 3-5 teachers	B3 paper prototype walkthrough
What devices exist per class?	Computer teachers, IT	Device audit
Do schools want coding despite CBSE's message?	Coordinators, computer teachers	Coordinator Q9; H10
Which language mix first?	Parents, teachers	Survey in 2-3 states
Coding, AI literacy or school-grade support for parents?	Parents	Indian parent survey (none exists)
Does CISCE or a state board adopt Class 3 CT&AI?	CISCE, SCERT	Track circulars; ICSE heads
Does placement + Express work in mixed classes?	Pilot students	Run in 2-3 Class 6-7 sections

9. Action plan

Tick actions as you finish them, add an owner and change the status. Due dates are pre-filled and editable.

> Fields are live: click a box to tick, type in Owner and Due, pick a Status. Save the file to keep your entries (outside the public repository).

A. This week: get ready (9-13 Nov)

Done	ID	Action	Owner	Due	Status
☐	A1	Fill [your number] / [your email] in the intro letter; decide on a Hindi version	Anoop	9 Nov	Not started
☐	A2	Create a private notes folder outside the repo and a backup location	Anoop	9 Nov	Not started
☐	A3	Print 20 copies of the B3 lesson (teacher pages 1-3, student page 4) and 20 device audit sheets	Anoop	11 Nov	Not started
☐	A4	Copy the tracking sheet to a private file; shortlist 25-30 schools matching the sample mix	Anoop	13 Nov	Not started
☐	A5	Send intro letters to the first 10 schools via networks and sahodaya complexes	Anoop	13 Nov	Not started

B. Interviews (16 Nov - 11 Dec)

Done	ID	Action	Owner	Due	Status
☐	B1	Run ~6 schools a week; 2-4 roles per school	Anoop	11 Dec	Not started
☐	B2	Fill the interview note form on the same day as each interview	Anoop	ongoing	Not started
☐	B3	Update the hypothesis tracker after every 5 schools	Anoop	every 5 schools	Not started
☐	B4	Ask every coordinator about coding plans despite CBSE's "no coding" message (Q9)	Anoop	every school	Not started
☐	B5	Ask whether CBSE's project list / rubrics or AI handouts have reached the school (Q8)	Anoop	every school	Not started
☐	B6	Do a device audit wherever there is a computer teacher; check Class 8 lab readiness	Anoop	every school	Not started

C. Decide (14-18 Dec)

Done	ID	Action	Owner	Due	Status
☐	C1	Group quotes by theme per fee tier and board; write the 2-page findings memo	Anoop	16 Dec	Not started
☐	C2	Apply the decision rule: go / no-go on a December pilot offer	Anoop	18 Dec	Not started
☐	C3	If go: send pilot offers and collect letters of intent before book lists close	Anoop	Jan 2027	Not started

D. Close research gaps

Done	ID	Action	Owner	Due	Status
☐	D1	Watch the CBSE webinar recording for anything not on the slides	Anoop	20 Nov	Not started
☐	D2	Read NCF-SE 2023 directly for the CT goal wording	Anoop	30 Nov	Not started
☐	D3	Get CISCE Classes I-VIII Computer Studies guidelines and AI 417 / IT 402 syllabi	Anoop	30 Nov	Not started
☐	D4	Open the handbook links: Google Docs worksheets, YouTube videos, Class 8 cat/dog dataset	Anoop	30 Nov	Not started
☐	D5	Collect UDISE+ 2025-26 device and Class 3-8 enrolment data for a bottom-up market model	Anoop	Dec	Not started
☐	D6	Legal review of DPDP Act child-data consent obligations	Anoop	before pilot	Not started
☐	D7	Check Google Play ratings for PictoBlox, CuriousJr and Codingal	Anoop	Dec	Not started

E. Monitor CBSE

Done	ID	Action	Owner	Due	Status
☐	E1	Check the CT&AI page monthly for the project list with rubrics and Concept Handouts	Anoop	monthly	Not started
☐	E2	Track circulars on Class 9 CT&AI for 2027-28 and any CISCE/state adoption	Anoop	monthly	Not started

F. Only after a "go" decision

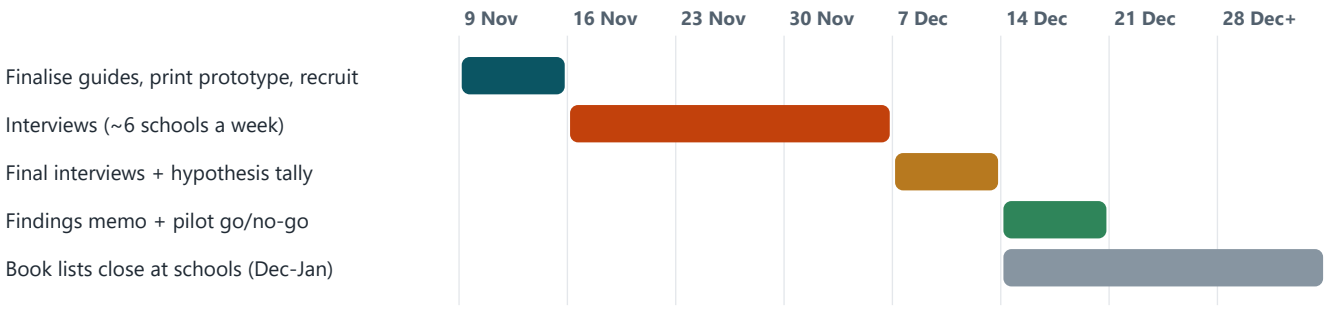
Done	ID	Action	Owner	Due	Status
☐	F1	Map every NCERT maths chapter (Classes 3-8) to CT skills and candidate levels	Anoop	Jan 2027	Not started
☐	F2	Draft one project kit per class, aligned with CBSE's example topics	Anoop	Feb 2027	Not started
☐	F3	Design a device-light, projector version of the Class 8 Teachable Machine labs	Anoop	Feb 2027	Not started
☐	F4	Write teacher scripts and slides for the Class 6 AI chapters	Anoop	Mar 2027	Not started

G. Go-to-market (added 6 Nov 2026)

Done	ID	Action	Owner	Due	Status
☐	G1	Apply the interview-guide changes and add H11-H13 (done in this version)	Anoop	13 Nov	Not started
☐	G2	Check the current CBSE bye-laws text on supplementary and digital material	Anoop	20 Nov	Not started
☐	G3	Get samples of Oxford, Goyal, Tekie.AI, Thinking Juggernaut and PiyushAI materials	Anoop	27 Nov	Not started
☐	G4	Shortlist 3-5 coined brand names; trademark attorney search (Classes 9, 16, 28, 41, 42)	Anoop	4 Dec	Not started
☐	G5	Tax adviser: GST on a printed workbook bundled with an access code	Anoop	4 Dec	Not started
☐	G6	Confirm specimen and booklist deadlines with interviewed schools	Anoop	11 Dec	Not started
☐	G7	Data-protection lawyer: the six DPDP questions and a school data processing agreement	Anoop	15 Jan 2027	Not started

10. School interview plan

Validate the research with 15-20 schools before any design or code. Interviews finish by mid-December, while schools are still choosing next year's resources.



10.1 Hypotheses (H1-H13)

#	Hypothesis	Supported if	Kill signal
H1	CBSE schools are implementing CT&AI for Classes 3-8 in 2026-27	Most have timetabled it, at least for some classes	Most treat "advised" as optional and did nothing
H2	Schools struggle to deliver it with the teachers they have	Coordinators name teacher capacity, planning or assessment	Free CBSE books are enough
H3	Principals will pay for a product alongside the free books	They name a budget line and price range, or accept an anchor	"We only use free material" in target tiers
H4	A Rs 300-600 package is acceptable (Rs 449/599 booklist pack or Rs 299 + GST licence)	Most mid/premium ceilings at or above Rs 300 for a package	Ceilings consistently below Rs 300
H5	Unplugged-first lessons with optional devices fit real equipment	Shared lab or projector only	Every school has 1:1 devices
H6	A maths teacher can run one level in one period with a script	B3 walkthrough fits a period with minor questions	Can't run it without a CS teacher
H7	Assessment and reporting are a reason to buy	Coordinators want dashboards and report-card grades	Schools don't assess CT&AI
H8	ICSE schools have a similar need for Classes 1-8	ICSE heads describe a gap or want CT&AI alignment	Happy with publisher textbook
H9	Schools want a Scratch-to-Python route for Classes 6-10	Students reach Class 11 Python unprepared; demand for earlier Python	Content with Scratch until Class 11
H10	What CBSE didn't provide is what schools struggle with: coding, projects, hands-on AI, journal	Named as unsolved, unprompted	Free handbooks cover their needs
H11	Schools will put a CT&AI workbook with app access on the booklist	Principals name the booklist as the route	New booklist items blocked and no school budget
H12	Schools will replace their current Computers book with a combined pack	Principals name the book they would drop	Locked into a multi-year series
H13	Student-data handling matters to buyers	Coordinators raise data or consent unprompted	Nobody cares; compliance is a cost, not a selling point

10.2 Who to interview

Dimension	Mix (15-20 schools, 40-60 interviews)
Board	About 12 CBSE, 4-5 ICSE, 1-2 state board (Kerala best comparison)
Fee tier	Budget (< Rs 30,000), mid (Rs 30,000-1,00,000), premium (> Rs 1,00,000): about a third each
Location	At least 3 states; metro, tier 2 and tier 3
School type	Mostly standalone private; 2-3 chain schools
Language	At least 4 schools teaching mainly in Hindi or a regional language

Role	Why	Length
Principal / owner	Budget, decision, price	30 min
CBSE / academic coordinator	Implementation, timetable, assessment	30-40 min
Class 3-5 maths or EVS teacher	Who actually teaches CT in 3-5; paper prototype	40 min
Computer teacher	Devices, labs, Classes 6-8 AI and projects	30 min
Parents (optional)	Home practice, AI vs coding priorities	15 min

Recruiting: personal and alumni networks → school associations and sahodaya complexes → LinkedIn outreach to coordinators. Offer a free cross-school summary; don't offer discounts on a product that doesn't exist.

> **Ground rules**
>
> - **Learn, don't sell.** No pitch until the last five minutes, and only if asked.
> - **Past behaviour, not hypotheticals.** "What did you buy last year?" beats "Would you buy...?"
> - **Consent first.** Verbal consent for notes; permission before recording; school won't be named.
> - **No children.** No student interviews or recordings; observation needs written permission.
> - **Record evidence.** Direct quotes, attributed by role, not name.

10.3 Interview guides

Principal or owner (30 min)

1. Tell me about the school: board, students, classes, fees roughly.
2. Who decides which books and digital products the school buys, and when? **Who signs the website book list; does a fee committee review it?**
3. How are you handling CBSE's CT&AI curriculum this year? (ICSE: how do you teach computers in Classes 1-8?)
4. **Which Computers and CT&AI books are on your 2026-27 list, from which publisher, at what price? Did you consider Oxford, Goyal or Uolo's Tekie?**
5. What's working, and what isn't?
6. Walk me through the last time you bought a new academic product.
7. When do your book lists close? **How do publishers work with you (specimens, training, terms), and by when must specimens arrive?** Don't ask about commissions.
8. **Which fee heads do you charge (computer, smart class, activity)? Would a new product go on the book list, an existing fee, or the school budget? Has a fee committee ever questioned a charge?**
9. **Do parents buy books from the school, a named shop, or anywhere?**
10. **Price anchors (packages):** "a Rs 300 licence plus GST, a Rs 450-600 workbook with app access on the book list, or an Rs 800 book-plus-platform with training: easy, hard or impossible?" Then: **would it replace your current computer book?**
11. If you could fix one thing about how computing or CT&AI is taught here, what would it be?
12. Who else should I talk to? Open to a free pilot in one or two sections?

CBSE / academic coordinator (30-40 min)

1. Has CT&AI been timetabled for Classes 3-8? Which classes, how many periods?
2. Who teaches it in Classes 3-5? In Classes 6-8?
3. Are you using the free CBSE books? How are teachers finding them?
4. Did teachers attend CBSE training (district/regional workshops, lead-school sessions, DD PM e-Vidya videos)? Useful?
5. How is CT&AI assessed? **CBSE hasn't published marks weightage or a report-card format. What will this year's report card show?**
6. What do parents ask about it?
7. What would make it easier: lesson plans, worksheets, a platform, training, assessment tools?
8. Classes 6-8 have 40 hours of projects and the books have no materials. What are students doing? Who designs and marks them? **Has CBSE's promised project list with rubrics arrived?**
9. **CBSE's webinar says "no coding/software". Will the school teach Scratch or any coding in Classes 3-8 anyway? Why or why not?**
10. How are teachers keeping the Teacher Observation Journal? Is there a format?
11. Have vendors approached you about CT&AI (**Oxford, Goyal, Uolo Tekie, STEMpedia, LEAD, Thinking Juggernaut, PiyushAI, Scaleopal**)? What did they offer, at what price?
12. **When an app needs student logins, who creates them and what does the school share? Does the school take parent consent for apps? Has anyone raised the DPDP Act?**
13. (ICSE) Anything from CISCE about CT or AI for younger classes?

Class 3-5 maths or EVS teacher (40 min)

1. How long have you been teaching, which subjects?
2. Have you taught CT&AI lessons yet? Tell me about the last one.
3. What was hardest? Are the handbook puzzles too easy, about right, too hard? Have you tried the activities (Caesar cipher, Coin Flip Magic, Sorting Network, weighing-scale search)?
4. How much time do you have to plan a new lesson each week?
5. Have you used computer or tablet activities with your class? What happened?
6. **Paper prototype (20 min):** walk through the B3 lesson, then: Could you run this tomorrow? Hardest step? What would you change? Projector, paper or lab?
7. What would make you look forward to teaching this?

Computer teacher (30 min)

1. **Device audit** (use the audit form): devices, lab sharing, internet, projectors, OS and age.
2. What do you teach in Classes 3-8 today, with which tools?
3. CBSE gives you the AI literacy part and project assessment in Classes 6-8. Doing it this year? With which tools? **The Class 8 book has three Teachable Machine labs: has anyone run them? Do you have webcams, microphones and internet?**
4. When do students first write Python? CBSE has none before Class 11. How do students cope with Class 11 Python?
5. What do students find hardest moving from blocks to typing code?
6. **How do students log in to any platform today: roll number, school email or a parent's phone?**

Parents (optional, 15 min)

1. What does your child do in computer or CT&AI class? How do you know?
2. What do you want them to learn about computers and AI?
3. Have you paid for coding classes or apps outside school? Why stop or continue?
4. If the school program had home practice, how much would you pay a month?
5. What would make you trust an education app? **Comfortable with the school giving an app your child's class and roll number? Do you use DigiLocker?**

6. **Where did you buy this year's books?**

10.4 Analysis and decision rule

- Fill the note form the same day.
- Every 5 schools, update the hypothesis tracker.
- After all interviews, group quotes by theme (affinity mapping) separately per fee tier and board.
- Write a 2-page memo: what we learned, which hypotheses held, what changes, and whether to make a pilot offer.

Decision rule

Go ahead with a December pilot offer if H1, H2 and H3 are supported in at least half of the CBSE schools, H4 holds in the mid and premium tiers, and H11 (booklist route) holds in at least half of mid- and premium-tier schools. If H1 fails, the product shouldn't depend on the CBSE circular: revisit the parent channel and the ICSE and state-board angles.

Week	Dates (2026)	Work
1	9-13 Nov	Finalise guides, print the B3 prototype, start recruiting
2-4	16 Nov - 4 Dec	Interviews: about 6 schools a week
5	7-11 Dec	Final interviews, tally hypotheses
6	14-18 Dec	Findings memo and go/no-go on the pilot offer

11. Hypothesis tracker

Update after every 5 schools: count schools where the evidence was For, Against or Unclear, then set a running verdict.

#	Hypothesis	For	Against	Unclear	Verdict	Key quote / note
H1	CBSE schools are implementing CT&AI for Classes 3-8 in 2026-27				Too early	
H2	Schools struggle to deliver it with the teachers they have				Too early	
H3	Principals will pay for a product alongside the free books				Too early	
H4	A Rs 300-600 package is acceptable (Rs 449/599 booklist pack or Rs 299 + GST licence)				Too early	
H5	Unplugged-first lessons with optional devices fit real equipment				Too early	
H6	A maths teacher can run one level in one period with a script				Too early	
H7	Assessment and reporting are a reason to buy				Too early	
H8	ICSE schools have a similar need for Classes 1-8				Too early	
H9	Schools want a Scratch-to-Python route for Classes 6-10				Too early	
H10	What CBSE didn't provide is what schools struggle with: coding, projects, hands-on AI, journal				Too early	
H11	Schools will put a CT&AI workbook with app access on the booklist				Too early	
H12	Schools will replace their current Computers book with a combined pack				Too early	
H13	Student-data handling matters to buyers				Too early	

Schools interviewed so far	
CBSE schools where H1+H2+H3 supported	
H4 holds in mid and premium tiers?	
Running decision	Too early to call

12. Interview materials

Ready to use: intro letter, opening script, the Level B3 paper lesson, and fillable forms for notes and the device audit.

12.1 Introduction letter (one page)

Dear [Principal / Coordinator name],

I'm **Anoop Dobhal**, and I'm researching how schools are teaching **Computational Thinking and AI** in Classes 3-8, including CBSE's new curriculum for 2026-27. Before building anything, I want to understand what schools and teachers actually need.

I'm requesting short conversations with up to four people at [School name]: the principal or owner (30 min), the academic or CBSE coordinator (30-40 min), a Class 3-5 maths or EVS teacher (40 min, including feedback on one printed sample lesson) and the computer teacher, if any (30 min). In person, by phone or by video, between **16 November and 11 December 2026**.

What I'll ask about: how CT&AI and computer classes are going, what teachers find hard, what equipment is available, and how the school chooses resources. **What this isn't:** a sales visit. There's no product to buy yet.

Your privacy: the school and staff won't be named in anything I share; notes only, audio only with agreement; I won't talk to, photograph or record any students; you can skip any question or stop at any time.

What you get: a free summary of what I learn from 15-20 schools, sent in January 2027.

Phone / WhatsApp: [your number] · Email: [your email]

Warm regards, Anoop Dobhal, CodeSchool

12.2 Opening script (read aloud)

"*Thank you for your time. I'm researching how schools teach computational thinking and AI, so that anything we build later actually helps teachers and students. This isn't a sales call. I'll take notes. Is it all right if I also record the audio, just for my own notes? It won't be shared. Your name and the school's name won't appear in anything I share. You can skip any question or stop at any time. Do you have any questions before we start?*"

12.3 Level B3: Robo makes choices (paper prototype)

Band / level	B (Builders, Classes 3-5) · Level B3: conditionals
Written for	Class 4 (works for Classes 3-5) · one 40-minute period · no devices (projector optional)
Structure	PRIMM + Parsons: Predict & Run → Investigate → Modify → Parsons → Make
Learning goal	Explain that an **if-else** block makes the program choose between two actions, checked every time the loop repeats
Curriculum link	Class 4 *Hide and Seek* and *The Cleanest Village*; Class 3 skip rule (*Filling and Lifting*). Explicit if-then is a CBSE Class 7 outcome, so B3 is a deliberate stretch the interviews test

Step 1. Predict and run (8 min)

```
   1          2          3          4          5
[ START ]  [ PUDDLE ]  [ plain ]  [ PUDDLE ]  [ plain ]
  Robo >

when green flag clicked
repeat 4 times
    move forward 1 square
    if < on a puddle? > then
        say [Splash!]
    else
        say [Step]
    end
end
```

Pairs write what Robo will say before running it. One student acts as Robo on the floor grid. **Answer:** Splash! · Step · Splash! · Step.

Step 2. Investigate (8 min)

#	Question	Answer	Listen for
1	How many times did Robo say "Splash!"? Why?	2: on a puddle twice	Linking output to squares
2	Which block makes Robo choose?	The if-else block	Not pointing to "repeat"
3	If every square were plain?	Step × 4	Else runs when the condition is false
4	If the if-else were after (outside) the repeat?	Checks once, on square 5: "Step"	**Hardest:** checking every time needs the loop

Misconceptions: "both branches run" (point to if… then… else); "the if checks once at the start" (act it out square by square).

Steps 3-5: modify, Parsons, make (22 min)

- **Modify (8 min):** new row with STAR squares; change only the if-else so Robo says "Star!". Output: Step · Star! · Star! · Step. Stretch: jump instead of speaking.
- **Parsons (8 min):** reorder jumbled blocks so Robo reaches GOAL without hitting a wall; `turn left` is the trick block. Answer: `repeat 6 times / if wall ahead then turn right else move forward`.
- **Make (6 min):** pairs draw five squares and write "If Robo is on a _ square, then _, else _"; swap and predict.
- **Exit ticket (2 min):** `[START][puddle][puddle][plain]`, repeat 3 → Splash! · Splash! · Step; circle the choosing block.

Full teacher script and student sheet: level-b3-paper-prototype.md.

12.4 B3 observation sheet (fillable)

School code · teacher role · date

Step	Planned	Teacher's estimate	Questions or hesitations (quote)	Would change it?
1. Predict and run	8 min			-
2. Investigate	8 min			-
3. Modify	8 min			-
4. Parsons puzzle	8 min			-
5. Make	6 min			-
Exit ticket	2 min			-

1. Could you run this in one period tomorrow? What would stop you?

2. Which step would your students find hardest?

3. What would you change?

4. Projector, paper or computer lab?

5. Is the answer key enough, or would you need more help? What kind?

6. Which maths chapter could this sit inside? Is if-else too early for Class 4?

Teacher could run it without help -

Estimated total time -

12.5 Interview note form (fillable)

> One form per interview. Use **school codes (S01-S20)** and roles, never names. Save filled copies outside the repository.

School code · role · date

Board / fee tier / city type / medium

Interviewer Anoop Dobhal

Audio recorded (with consent) ☐ **CT&AI timetabled** - **Device summary** -

Key quotes

Current product and cost

Book list closes

Price-anchor reaction Rs 300 licence - Rs 450-600 pack - Rs 800 platform -

#	Hypothesis (short)	Evidence	Notes / quote
H1	CBSE schools are implementing CT&AI for Classes 3-8 in 2026-27	Not discussed	
H2	Schools struggle to deliver it with the teachers they have	Not discussed	
H3	Principals will pay for a product alongside the free books	Not discussed	
H4	A Rs 300-600 package is acceptable (Rs 449/599 booklist pack or Rs 299 + GST licence)	Not discussed	
H5	Unplugged-first lessons with optional devices fit real equipment	Not discussed	
H6	A maths teacher can run one level in one period with a script	Not discussed	
H7	Assessment and reporting are a reason to buy	Not discussed	
H8	ICSE schools have a similar need for Classes 1-8	Not discussed	
H9	Schools want a Scratch-to-Python route for Classes 6-10	Not discussed	
H10	What CBSE didn't provide is what schools struggle with: coding, projects, hands-on AI, journal	Not discussed	
H11	Schools will put a CT&AI workbook with app access on the booklist	Not discussed	
H12	Schools will replace their current Computers book with a combined pack	Not discussed	
H13	Student-data handling matters to buyers	Not discussed	

Surprises

Follow-ups	

12.6 Device audit (fillable)

Fill with the computer teacher or IT staff, one per school. Write **numbers**, not impressions; write "don't know" rather than guessing. Ask to see the lab.

School code · date · filled with (role)	

1. Computer lab

Number of computer labs	
Computers in main lab (total)	
…of which working today	
Operating system	
Approx. age of most computers (years)	
Typical RAM	
Students in a typical section	
Students per working computer in a lab period	
Sections sharing the lab	
Lab minutes/week: Class 4 section	
Lab minutes/week: Class 7 section	
Scratch / PictoBlox installed? Online or offline?	
Python installed?	

2. Tablets and other devices

School-owned tablets	
…Android / iPad / other	
…working today	
Which classes can use them?	
Do students bring phones/tablets for classwork?	

3. Classrooms

Classrooms (Classes 3-8)	
…with projector or smart board	
…working today	
…with teacher laptop/PC connected	
Projector in Class 3-5 rooms?	

4. Internet and Class 8 labs

Internet in lab (yes / no / unreliable)	
Internet in classrooms	
Connection type	
How often does it fail in a lesson?	
Filter? Is scratch.mit.edu / teachablemachine allowed?	
Webcams available (number)	
Microphones / headsets available (number)	

5. Power, support and funding

Power cuts in school hours	
Backup power for lab	
Who fixes devices, how long?	
Lab funded by a scheme (Samagra Shiksha ICT, ATL)?	
Atal Tinkering Lab? In use?	
Devices planned for next session?	

Interviewer's summary	-
Notes	

12.7 Tracking sheet (columns)

Use tracking-sheet-template.csv for S01-S20; copy it to a **private** file before adding contacts. Better: use the Research Tracker workbook, which logs H1-H13 evidence and checks the decision rule automatically.

13. Appendix

Document index, key sources and glossary.

13.1 Documents in the repository

Document	What it contains
README.md	Project overview and links to all research
Kids coding curriculum and market	Main research report (sections 3-5, 8)
Research notes	India curriculum, global curricula, learning progression, market landscape
cbse-verification.md	CBSE claims checked against official documents (section 6)
cbse-handbook-review.md	All 12 handbooks + webinar (section 7)
school-interview-plan.md	Hypotheses, sample, guides, analysis, timeline (section 10)
interview-materials/	Intro letter, B3 lesson, device audit, tracking sheet (section 12)

13.2 Key official sources

Source	Link
CBSE CT&AI page (all handbooks)	cbseacademic.nic.in/ct-ai.html
CBSE CT&AI curriculum, Classes 3-8 (2026-27)	cbseacademic.nic.in/web_material/CurriculumMain27/CTAI_Pri_2026-27.pdf
Circular Acad-15/2026: launch	cbseacademic.nic.in/web_material/Circulars/2026/15_Circular_2026.pdf
Circular Acad-18/2026: resource books	cbseacademic.nic.in/web_material/Circulars/2026/18_Circular_2026.pdf
Circular Acad-70/2026: Class 9 scheme	cbseacademic.nic.in/web_material/Circulars/2026/70_Circular_2026.pdf
CBSE webinar slides, 28 April 2026	cbseacademic.nic.in/web_material/CurriculumMain27/CTAI_WebinarPPT_2026...
CBSE webinar recording	youtube.com/live/KXYvvXZtYyk
Computer Applications 165, Class IX	cbseacademic.nic.in/web_material/CurriculumMain27/SecPart1/Computer_Ap...
Computer Science 083	cbseacademic.nic.in/web_material/CurriculumMain27/SecPart2/Computer_Sc...
Informatics Practices 065	cbseacademic.nic.in/web_material/CurriculumMain27/SecPart2/Informatics...
NEP 2020	static.pib.gov.in/WriteReadData/userfiles/NEP_Final_English_0.pdf
Samagra Shiksha ICT (DoSEL)	dsel.education.gov.in/scheme/ict-samagra
CSTA PK-12 Standards (2026)	csteachers.org/pk12standards/
England computing programmes of study	www.gov.uk/government/publications/national-curriculum-in-england-comp...
Costa 2023: game concepts in programming learning	repositorio.iscte-iul.pt/bitstream/10071/29213/1/article_95533.pdf
Sentance, Waite & Kallia 2019: PRIMM	core.ac.uk/download/237088209.pdf
CS Unplugged	www.csunplugged.org/
Teachable Machine	teachablemachine.withgoogle.com

The main report has the full list of 100+ labelled sources, including news and secondary items.

13.3 Glossary

Term	Meaning
CT	Computational thinking: decomposition, pattern recognition, abstraction, algorithms, data, troubleshooting
CT&AI	CBSE's Computational Thinking and Artificial Intelligence curriculum for Classes 3-8 (2026-27)
TWAU	The World Around Us, the NCERT environmental studies subject in Classes 3-5
SH / TH	Student Handbook / Teacher Handbook
NCF-SE	National Curriculum Framework for School Education, 2023
PRIMM	Predict, Run, Investigate, Modify, Make: a lesson structure for programming
Parsons problem	Put jumbled code blocks in the right order
Band / level	CodeSchool's proposed multi-class bands (A-E) made of mastery-gated levels
Express track	A fast catch-up route for older beginners
B2B2C	Sell to schools, reach parents through the school
GeM	Government e-Marketplace, India's public procurement portal
CPD	Continuing professional development for teachers
DPDP Act	India's Digital Personal Data Protection Act, 2023
LOI	Letter of intent